Learning for Data Science

Course 9: Hypothesis Testing

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Outline

How to Compare Two Distributions fairly?

T-Test

What if There are Multiple Groups

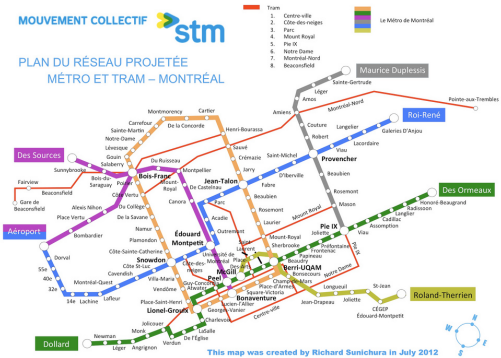
Post Hoc Analysis

What about Categorical Data?

How to Compare Two Distributions fairly?

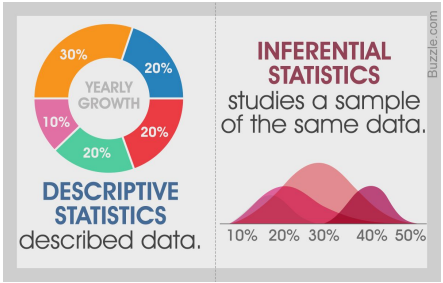
- Problem: I want to figure out how to get to class on time
- My choices are take the bus, or take the metro
- Which method is more likely to get to class on time?

- So far I have collected 30 samples
  - 10 positives for metro
  - 5 positives for the bus
- Should I take the bus?
  - How sure of this am I?



Inferential Statistics?

- There is rarely enough information to be sure something is true
- Inferential Statistics: Collect some data, and then infer something about the true distribution that was sampled.



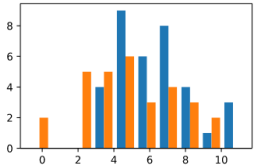
What is a Hypothesis?

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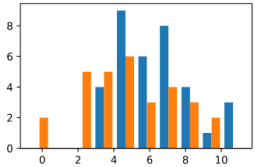
- Consider the question "Do students in Section A get to class faster then section B."
  - However, you are a student and do not have access to everyone's schedule.
  - You can randomly sample queries from 20% of the class
  - Class A time mean 4.85
  - Class B time mean: 6.25



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- Consider the question "Do students in Section A get to class faster then section B."
  - However, you are a student and do not have access to everyone's schedule.
  - You can randomly sample queries from 20% of the class
  - Class A time mean 4.85
  - Class B time mean: 6.25
  - It looks like class A is faster.
    - However, you did not sample all students in the class.
  - Hypothesis Testing: Check how likely this result is to occur randomly using inferential statistics.



Types of Hypothesis

- **Null hypothesis** (H_0): Nothing was found.
- **Alternative hypothesis** (H_1): The alternative option being studied to see if it is true.
- In the class comparison example.
  - The Null hypothesis is, there is not enough information to conclude anything
  - The Alternative hypothesis: There is enough information to say something is possible (with some confidence).

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- Wording the hypothesis for asking a question
  - H_0 — There is no difference between means in class A and those in B.
  - H_1 — There is a difference between means of in class A and those B.

T-Test

- Consider a situation with:
- Two distributions with normal distributions,
- ... with equal variances.
- Want to conclude that the two distributions have different means.
- Reminder: The true generating distribution is unknown.
- How it works
  - Considering that sampling is a random process
  - What is the probability of getting the same result again?

T-Test Analysis

$$t = \frac{\overbrace{\bar{x}_1 - \bar{x}_2}^{\text{signal}}}{\underbrace{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}_{\text{noise}}}$$

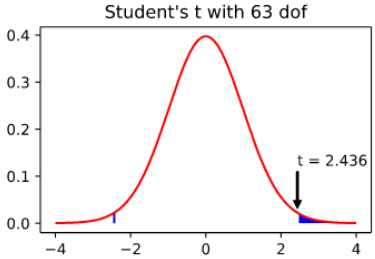
$$\frac{\text{mean difference}}{\text{total variance of two groups}} = \frac{\text{signal}}{\text{noise}}$$

Where:

- $\bar{x}_1, \bar{x}_2$ – sample means generated from 1st and 2nd group respectively;
- n_1, n_2 – sample size of 1st and 2nd group respectively;
- $s_p = \sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}}$ – is also called "*pooled standard deviation*"
- s_1, s_2 = standard deviation of 1st and 2nd group respectively.

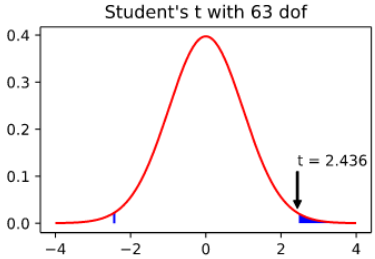
T-Test Analysis

- The Class A/B samples have t-stat $= 2.436$
- 0.00885 is area under curve, right of 2.436.
- 0.00885 is only right. There is a total of $2 \times 0.00885 = 0.0177 = 1.77\%$ probability of being farther than 2.436 from zero.



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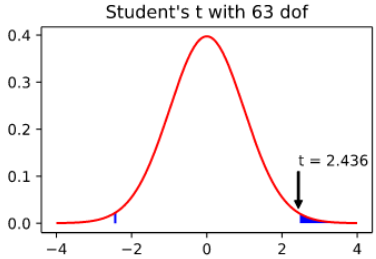
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- Which means, the probability of getting these samples from (normally-distributed, equal variance) populations with identical means is 0.0177.

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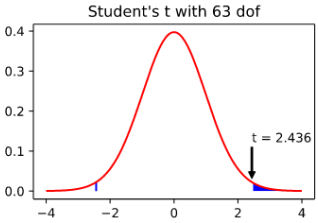
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- Which means, the probability of getting these samples from (normally-distributed, equal variance) populations with identical means is 0.0177.
- Conclusion: it's very unlikely that H_0 is true.

Testing Normality

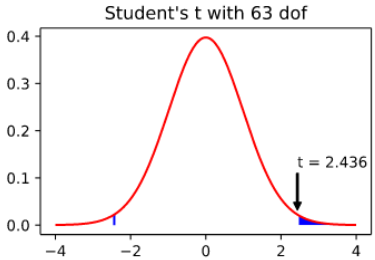
- It is possible the assumptions for t-test are not true
- Again: The true distribution is unknown



- What can we do? The same thing we do every day
  - Use a normality test (e.g. Shapiro–Wilk) and ask "Does the data look Gaussian?"
  - This is a backwards test
  - The goal is $p > 0.05$

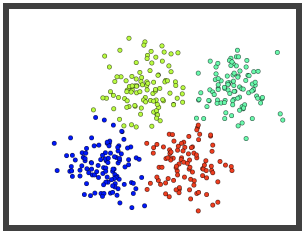
Testing Are the Variances the Same?

- **Again: The true distribution is unknown**
- Given the available data, how likely is it the variances are the same?
  - Called Homoscedastic
- **Levene**'s test has that the two samples have equal variance.
  - With small p, that assumption is rejected
- There is a version of the T-test that does not assume equal variance.
  - **Welch's t-test**



What if There are Multiple Groups

- T-test are designed to work with two distributions
- What if there are three sets of samples, and the question is whether any are different? E.g., is there any difference in height between Bus, Bike, and walk?

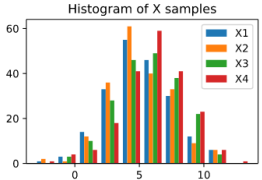


- One option: do a T-test for Bus vs Metro; Bus vs Bikes; Metro vs Bike. Ask if there are any differences with $p < 0.05$.
  - But remember what $p < 0.05$ means: there is a 5% probability of error rejecting the null incorrectly, **just by chance**.
  - If three T-tests are done, then the probability of no incorrect rejection of the null is: $0.95^3 = 0.86$
- The effective p is now 0.14: much less confidence in the results.

ANOVA (ANalysis Of VAriance)

- A test to determine if the means of any of the groups differ. Similar to a T-test, but for >2 groups.

- Assumptions:
  - Observations independent and identically distributed (iid).
  - Groups are normally distributed.
  - Groups have equal variance.

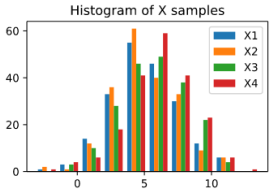


- Are any of the means different?
  - Hard to tell
- Great, but unsatisfying. Some groups have different means, but the test does not say which ones.

ANOVA (analysis of variance)

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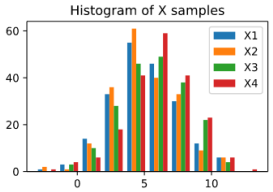
- Are any of the averages different?
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ANOVA (analysis of variance)

- A test: determine if means of any groups differ. Similar to a T test, but for groups > 2.

- Are any of the averages different?
- Hard to say… the test says yes!
- Great, but unsatisfying.



- Know there are groups with different averages, but which ones.

```
from scipy import stats
anova = stats.f_oneway(x1, x2, x3, x4)
print(anova)
print(anova.pvalue)
```

```
F_onewayResult(statistic=6.301848031347299, pvalue=0.0003159232
0.0003159232567907352
```

Post Hoc Analysis

- A test to determine if the means of any of the groups differ. It's very much like a T-test, but for >2 groups.
- If you get significance in an ANOVA, you can then do post hoc analysis. That is, do pairwise comparisons between each variable.
- Must be done correctly, to avoid doing many separate tests and looking at invalid p-values.
- The most commonly-suggested test for this case is Tukey's HSD (Honest Significant Difference) test.

Tukey's HSD (Honest Significant Difference) test

- Gives a comparison of if the Means for different groups are similar.
- There are 4 groups x_1, x_2, x_3, x_4

Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
x1	x2	-0.0202	0.9	-0.6538	0.6135	False
x1	x3	0.4763	0.2143	-0.1574	1.1099	False
x1	x4	0.8956	0.0017	0.262	1.5293	True
x2	x3	0.4965	0.1824	-0.1372	1.1301	False
x2	x4	0.9158	0.0012	0.2822	1.5494	True
x3	x4	0.4194	0.3224	-0.2143	1.053	False

Hacking p-values

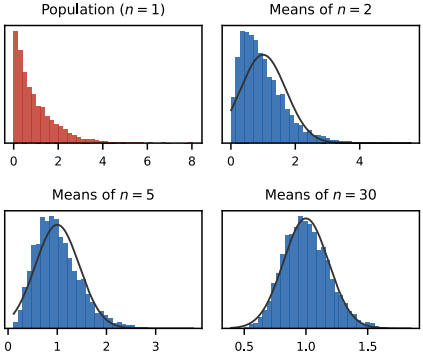
- **It is important not to trick oneself (or others) with statistics.**
  - Keep in mind what a p-value really is.
- Imagine a survey that looks at fitness by location. The survey collects several variables: height, weight, city where you live.
- Bad thing #1: collect data until the ANOVA of weight by city reaches $p < 0.05$.
- If you're designing the experiment to get the conclusion you want, then you will not receive true results.

Hacking p-values

- Don't try tests until one works...
- ANOVA of weight by city: $p < 0.05$.
- ANOVA of height by city: $p < 0.05$
- Weight T-test for Vancouver vs Toronto: $p < 0.05$
- Weight T-test for Vancouver vs Calgary: $p < 0.05$
- On each test, you have a 5% chance of rejecting the null hypothesis by chance.
- Tukey's HSD (Honest Significant Difference) takes this into account!

Central Limit Theorem

- Consider any probability distribution with mean μ and variance σ. Sample n values from X, and call it S_i. Keep doing that to generate S_1. Repeat this for S_2, S_3.
- The sample means, S_k, tend toward a normal distribution with large enough n.



Some Tips

- These tests work well if you have enough data
- The data should be normally distributed
- *40 point is okay to start*

What about Categorical Data?

- What if your data has even less structure and is categorical?
- **Chi-Squared test**
  - A chi-squared (test) works on categorical totals like this (assuming > 5 observations in each category).
  - Null hypothesis: The categories are independent. i.e. it doesn't matter what category you're in; the proportions will be the same.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

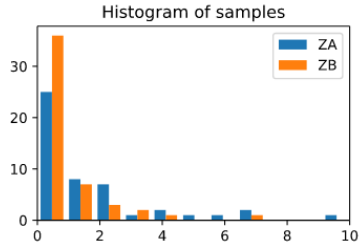
- O = the frequencies observed,
- E = the frequencies expected,
- $\sum$ = the "sum of"

Mann–Whitney U-test

- What if no assumptions can be made about the data distribution?
- **Mann–Whitney** U-test
  - This is a Non-parametric test
  - Can be used to decide that samples from one group are larger/smaller than another
- Assumptions:
  - Observations are independent.
  - Values are ordinal: can be sorted.

Mann–Whitney U-test

- H_0: if x, y are chosen from X, Y what is $P(x < y) = \frac{1}{2}$?
- H_1: $P(x < y) \neq \frac{1}{2}$?
  - Values from one group tend to sort higher than other
- Mann-Whitney doesn't care about magnitude of differences, only sort order.
- Corollary: If benchmarking code, run three times each, and if all three runs are faster/slower, then Mann-Whitney will give significance with $p = 0.04$.



Quiz/Poll

- You want to compare two groups, but the data is clearly not Gaussian. Which test should you use?
- (A) Student's t-test, (B) Welch's t-test, (C) Mann–Whitney U-test, or (D) ANOVA

Quiz/Poll

- How can you check that your data is normally distributed before a t-test?
- (A) Levene's test, (B) a Shapiro–Wilk test, (C) a chi-squared test of independence, or (D) Tukey's HSD test

References

- https://ggbaker.ca/data-science/slide-content/stats-tests.html#/title-2
- https://towardsdatascience.com/t-test-and-hypothesis-testing-explained-simply-1cff6358633e
- https://ggbaker.ca/data-science/slide-content/inferential.html#/title-6